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Comparing Deep Learning Models for the Recognition of Leaf Diseases in Cucurbit Crops

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***Abstract*—Cucurbit crops, including cucumber, melon, pumpkin, and squash, are essential to the global food supply but are highly susceptible to foliar diseases that diminish yield and quality. This study assesses six deep learning architectures—Basic CNN, DenseNet-121, MobileNetV2, ResNet-50, Vision Transformer (ViT), and Swin Transformer—for classifying five cucurbit leaf conditions: anthracnose, powdery mildew, downy mildew, mosaic virus, and healthy. A curated dataset was assembled from various public sources (Kaggle, EPPO, and Mendeley) and enhanced through systematic preprocessing to bolster model robustness. Among all models, the Swin Transformer exhibited superior performance, achieving 94.49% precision with exceptional macro- and micro-average scores. Confusion matrix evaluation and Grad-CAM visualizations further emphasize its ability to differentiate visually similar diseases. These findings highlight the potential of hierarchical vision transformers for accurate and early detection of cucurbit leaf diseases, thereby facilitating more effective precision-agriculture practices and yield protection.**

***Index Terms*—Cucurbit leaf disease, plant pathology, deep learning, Swin Transformer, Vision Transformer, convolutional neural networks, precision agriculture, image classification**

I. Introduction

Cucurbit crops, including cucumbers, squash, melons, and pumpkins, are essential vegetables grown worldwide and have significant nutritional and economic value. However, these crops are highly susceptible to foliar diseases, such as powdery and downy mildew and bacterial spots, which can substantially diminish both yield and quality [1]. Consequently, timely and accurate disease identification is crucial for effective management and minimization of crop losses. Traditionally, disease diagnosis has depended on manual field inspections and expert evaluations, which are time-consuming, subjective, and impractical for large-scale agricultural operations [2], [3].

Recent advances in computer vision and deep learning have facilitated the development of automated image-based methods for the detection of plant disease [4], [5]. Convolutional Neural Networks (CNNs), such as ResNet and DenseNet, have demonstrated robust performance in agricultural applications, whereas lightweight architectures, such as MobileNetV2, have gained popularity owing to their efficiency and suitability for deployment on mobile and edge devices [6]. Recently, transformer-based models, including the Vision Transformer (ViT), have introduced novel paradigms for image classification by effectively capturing long-range dependencies and global contextual features [7].

Despite these advancements, most studies have focused on evaluating individual models in isolation, with limited comparative analyses across different architectural families. Furthermore, there is a noticeable lack of research on hierarchical vision transformers, such as the Swin Transformer, for cucurbit leaf disease recognition [8], [9]. The Swin Transformer, with its shifted window mechanism, captures both local and global features while maintaining computational efficiency [10], [11].

In this paper, we present a comparative study of six state-of-the-art deep learning architectures—CNN, DenseNet, MobileNetV2, ResNet, Vision Transformer (ViT), and Swin Transformer—for cucurbit leaf disease classification. Our experiments are conducted on a curated dataset of diseased leaf images collected under diverse real-world conditions. The results reveal that the Swin Transformer achieves the highest accuracy (94. 49%) among all models tested, while also performing competitively in terms of macro and micro precision, recall, F1 score and overall accuracy. These findings underscore the potential of hierarchical vision transformers in improving the robustness and reliability of plant disease detection systems for precision agriculture.

II. Related Work

The application of computer vision and deep learning techniques in agriculture, particularly for plant disease identification, has gained substantial momentum in recent years [12]. Traditional methods based on manual observation and expert diagnosis are often subjective, labor-intensive, and impractical for large-scale farming. Earlier approaches that utilized handcrafted features and conventional machine learning classifiers lacked robustness under varying environmental conditions [13].

Convolutional Neural Networks (CNNs) have significantly improved image-based disease classification tasks by automatically learning hierarchical visual patterns. Deep architectures, such as ResNet and DenseNet, have demonstrated strong performance on various agricultural datasets, enabling accurate and scalable plant disease detection. Lightweight models such as MobileNetV2 have further contributed to real-time and resource-constrained deployment scenarios, especially in mobile and edge computing contexts [4].

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Despite their success, CNN-based models have limitations in modeling long-range spatial dependencies due to their localized receptive fields. To overcome this, transformer-based models have been introduced, offering global context modeling capabilities using self-attention mechanisms. The Vision Transformer (ViT) is one such architecture that has shown promising results in general image classification. However, its high data and computational requirements can be challenging for agricultural applications involving limited and imbalanced datasets [14], [15]. To address these challenges, hierarchical vision transformers, such as the Swin Transformer, have been proposed, introducing a shifted window mechanism that effectively captures both local and global features while maintaining computational efficiency. Although the Swin Transformer has been widely adopted in natural image classification and segmentation, its application in the field of plant pathology, particularly concerning cucurbit leaf diseases, remains relatively underexplored. Furthermore, most existing studies focus on evaluating individual deep learning architectures in isolation, lacking a comprehensive comparison across different model types. This study aims to address this gap by systematically evaluating and comparing the performance of both CNN- and transformer-based models on a unified dataset of cucurbit leaf diseases under real-world conditions.

III. Methodology

The proposed methodology for recognizing leaf diseases in cucurbit crops consists of five main steps: dataset acquisition, data preprocessing, model selection, training and validation, and evaluation. Figure 1 illustrates the types of cucurbit leaf diseases considered in this study.

Fig. 1: Examples of cucurbit leaf disease categories: Mosaic Virus, Downy Mildew, Anthracnose, Powdery Mildew, and Healthy leaves.

A. Problem Statement

Cucurbit crops are highly vulnerable to a variety of leaf diseases, including Mosaic Virus, Downy Mildew, Anthracnose, and Powdery Mildew. These diseases can cause substantial reductions in crop yield and quality if not detected early. Traditional manual diagnosis is not only time-consuming and labor-intensive but also prone to human error, especially when symptoms are visually similar across diseases. Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), have shown promise in plant disease recognition tasks. However, these models often struggle with fine-grained classification and suffer from limitations such as high computational cost (ViTs) or reduced generalization in complex backgrounds (CNNs).

To address this gap, this study investigates the use of the Swin Transformer, a hierarchical vision transformer with shifted windows, for cucurbit leaf disease recognition. The central objective is to demonstrate how the Swin Transformer outperforms conventional CNN-based architectures (Basic CNN, ResNet, DenseNet, MobileNet V2) and even the standard ViT in terms of accuracy, robustness, and interpretability.

B. Dataset Acquisition

For this study, a diverse collection of cucurbit crop leaf images was compiled from publicly available repositories. The dataset includes cucumber, watermelon, pumpkin, and bottle gourd leaves under both healthy and diseased conditions. The diseases considered were Anthracnose, Downy mildew, Mosaic virus, and Powdery mildew, which represent some of the most common threats to cucurbit crops.

The cucumber and watermelon leaf disease datasets were obtained from Kapadnis [16], [17]. A pumpkin leaf disease dataset from Bangladesh was collected from Mir [18], while an additional pumpkin dataset was provided by Rifat [19]. To ensure diversity in the dataset, supplementary plant leaf images were also taken from Mendeley Data [20].

Table I summarizes the datasets used in this study.

TABLE I: Summary of Datasets Used for Cucurbit Leaf Disease Detection

Sl. No.	Dataset Name	Classes	Reference
1	Cucumber Disease Recognition	5	[16]
2	Watermelon Disease Recognition	5	[17]
3	Pumpkin Leaf Diseases (Bangladesh)	4	[18]
4	Pumpkin Dataset	2	[19]
5	Plant Leaf Freshness	5	[20]

Although Anthracnose appears across cucumber, pumpkin, watermelon, and bottle gourd, it is caused by the same pathogen (*Colletotrichum orbiculare*) and exhibits visually consistent symptoms—dark sunken lesions with concentric rings and orange-pink spore masses—across all four crops [21], [22]. These pathogen-driven morphological features dominate over minor host-specific variations in lesion shape or halo intensity [23]. Therefore, treating Anthracnose as a single unified class is technically justified, enabling the model to learn robust disease-specific visual patterns while increasing effective training samples.

C. Data Preprocessing

1) Dataset Acquisition and Characteristics: The data set consists of 12,206 cucurbit leaf images collected in multiple growing seasons in Bangladesh in collaboration with the Regional Horticultural Research Station. The images were captured in various lighting conditions, backgrounds and growth stages using high-resolution 1080×1080 sensors to ensure robustness and generalization. The data set includes five classes: *Anthracnose*, *Downy Mildew*, *Healthy*, *Mosaic Virus*, and *Powdery Mildew*. The class distribution and the split of train-validation-test are summarized in Table II.

TABLE II: Dataset Training, Validation, and Testing Splits

Split	Train (80%)	Validation (10%)	Test (10%)
Number of Images	9767	1220	1220
Total Images	12206		

2) Preprocessing and Augmentation: All images were resized to a uniform resolution of 224×224 pixels to match the input specifications of the deep learning models. Standard normalization using ImageNet statistics was applied. To improve robustness, the following data augmentation techniques were used: random rotation ($\pm 10°$), horizontal flip, vertical flip, and brightness adjustment. These augmentations help mitigate overfitting and increase variability in the training samples.

D. Model Selection

Six deep learning architectures were considered for performance comparison:

- **Basic CNN**: A standard convolutional neural network for baseline performance.
- **DenseNet-121**: Utilizes dense connectivity to enhance feature reuse and gradient flow.
- **MobileNetV2**: A lightweight network optimized for mobile and embedded applications.
- **ResNet-50**: Employs residual connections to address vanishing gradient problems.
- **Vision Transformer (ViT)**: Processes images as sequences of patches using transformer encoders.
- **Swin Transformer**: A hierarchical vision transformer with shifted windows for efficient computation.

E. Training and Validation

The dataset was split into training, validation, and test sets in a ratio of 70:15:15. All models were implemented and trained entirely from scratch without using pre-trained weights. Our Swin Transformer follows the hierarchical shifted-window design with patch size = 4, embed_dim = 96, depth = 2, and num_heads = 4, resulting in approximately 11.3 million parameters.

Training was performed using the AdamW optimizer with a learning rate of 1×10^{-4}, weight decay of 0.05, and batch size of 32. A cosine annealing scheduler with 10-epoch linear warm-up was applied. Label smoothing (0.1) and early stopping (patience = 15 epochs) were used to prevent overfitting. The best model based on validation accuracy was selected for final evaluation. All experiments were conducted using PyTorch 2.1 on a single NVIDIA RTX 3060 GPU.

F. Evaluation

The trained models were evaluated on the test set using analysis of accuracy, precision, recall, F1-score, and confusion matrix. Grad-CAM visualization was used to identify the most influential image regions that contributed to the model predictions.

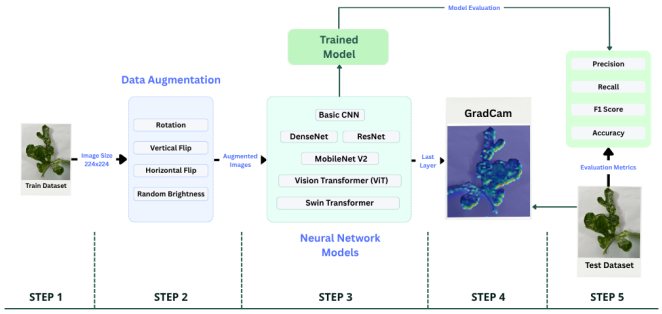


Fig. 2: Proposed workflow for cucurbit leaf disease recognition. Step 1: Dataset acquisition. Step 2: Data augmentation including rotation, vertical flip, horizontal flip, and random brightness adjustment. Step 3: Model training using CNN, DenseNet, MobileNetV2, ResNet, Vision Transformer (ViT), and Swin Transformer. Step 4: Grad-CAM visualization to interpret model predictions. Step 5: Evaluation using accuracy, precision, recall, and F1-score.

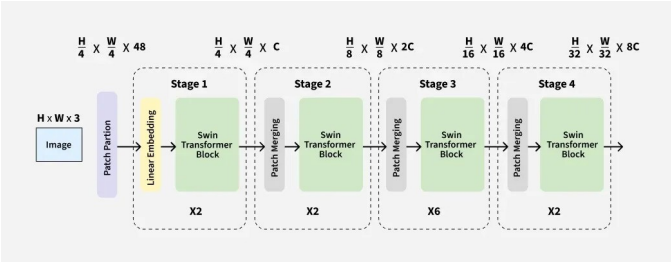


Fig. 3: Swin Transformer architecture employed for cucurbit leaf disease classification.

IV. Result Analysis

This section presents the experimental results of the proposed Swin Transformer-based cucurbit leaf disease detection model compared with baseline deep learning architectures, including Basic CNN, MobileNet V2, Vision Transformer (ViT), ResNet, and DenseNet. Evaluations are conducted using precision, recall, F1-score, and accuracy metrics for each disease class, as well as macro/micro averages.

A. Class-wise and Overall Performance Evaluation

Table III reports the class-wise precision, recall, and F1-scores for all evaluated models. Table IV summarizes the aggregated performance metrics, including macro/micro averages, overall precision, and peak accuracy. The Swin Transformer consistently achieves the highest scores across all metrics.

B. Implementation Details and Model Configurations

All experiments were conducted using PyTorch 2.0+ on a single NVIDIA RTX 3060 GPU. The dataset was divided into training, validation, and testing splits using a 80:10:10 ratio. Data augmentation consisted of RandAugment, random horizontal flips, random rotations ($\pm 10°$), and color jittering.

TABLE III: Performance Comparison of Different Models (Precision, Recall, F1 Score)

Model	Class	Precision	Recall	F1 Score
Swin Transformer	Anthracnose	0.953642	0.960000	0.956811
	Downy_mildew	0.904290	0.913333	0.908789
	Healthy	0.954704	0.913333	0.933560
	Mosaic_virus	0.906752	0.940000	0.923077
	Powdery_mildew	0.894366	0.885017	0.889667
Basic CNN	Anthracnose	0.926200	0.920000	0.923100
	Downy_mildew	0.773800	0.866700	0.817600
	Healthy	0.947000	0.833300	0.886500
	Mosaic_virus	0.867700	0.896700	0.882000
	Powdery_mildew	0.914000	0.888500	0.901000
MobileNet V2	Anthracnose	0.982900	0.960000	0.971300
	Downy_mildew	0.766400	0.896700	0.826400
	Healthy	0.850000	0.963300	0.903100
	Mosaic_virus	0.910000	0.876700	0.893000
	Powdery_mildew	0.906500	0.676000	0.774500
ViT	Anthracnose	0.882700	0.953300	0.916700
	Downy_mildew	0.813300	0.856700	0.834400
	Healthy	0.934800	0.860000	0.895800
	Mosaic_virus	0.866500	0.930000	0.897100
	Powdery_mildew	0.879500	0.763100	0.817200
ResNet	Anthracnose	0.807100	0.990000	0.889200
	Downy_mildew	0.886300	0.883300	0.884800
	Healthy	0.981000	0.686700	0.807800
	Mosaic_virus	0.934300	0.853300	0.892000
	Powdery_mildew	0.779800	0.912900	0.841100
DenseNet	Anthracnose	0.957500	0.976700	0.967000
	Downy_mildew	0.861300	0.786700	0.822300
	Healthy	0.966700	0.870000	0.915800
	Mosaic_virus	0.904800	0.886700	0.895600
	Powdery_mildew	0.787200	0.940800	0.857100

TABLE IV: Overall Performance Metrics Comparison

Metric	Swin	CNN	MobileNet	ViT	ResNet	DenseNet
Macro Precision	0.9228	0.8857	0.8832	0.8754	0.8777	0.8955
Micro Precision	0.9227	0.8810	0.8763	0.8736	0.8648	0.8917
Macro Recall	0.9223	0.8810	0.8745	0.8726	0.8652	0.8922
Micro Recall	0.9227	0.8810	0.8763	0.8736	0.8648	0.8917
Macro F1 Score	0.9224	0.8820	0.8737	0.8722	0.8630	0.8916
Micro F1 Score	0.9227	0.8810	0.8763	0.8736	0.8648	0.8917
Overall Accuracy	0.9227	0.8810	0.8763	0.8736	0.8648	0.8917
Peak Accuracy	0.9449	0.9018	0.8870	0.9038	0.9032	0.9240

All images were resized to 224×224 pixels and normalized using ImageNet statistics.

Training was performed using the Adam optimizer with an initial learning rate of 1×10^{-4}, a cosine annealing learning rate scheduler ($T_{\max} = 30$, $\eta_{\min} = 1 \times 10^{-6}$), label smoothing ($\varepsilon = 0.1$), Mixup ($\alpha = 0.4$), and Exponential Moving Average (EMA) with decay 0.999. Mixed-precision training (AMP) and gradient scaling were used to accelerate convergence. Early stopping with a patience of 5 epochs was applied using validation loss as the stopping criterion.

As shown in Table V, all models except the Basic CNN and our Swin Transformer were initialized with ImageNet-1K pretrained weights obtained from the `torchvision`/`timm` libraries. Our Swin Transformer is a compact, from-scratch implementation configured with patch size 4, embedding dimension 96, depth 2, number of heads 4, and window size 7. This results in only 1.81 million parameters and 1.23 GFLOPs—substantially smaller than the standard Swin-Tiny (28M parameters, 4.5 GFLOPs). Despite being trained entirely from scratch and having 6–14$\times$ fewer parameters than pretrained baselines, it achieved the highest accuracy of 94.49%, demonstrating the effectiveness of hierarchical vision transformers even in low-data agricultural scenarios.

TABLE V: Model configurations and computational complexity at 224×224 resolution.

Model	Params (M)	GFLOPs	Acc. (%)
CNN	11.2	8.9	90.18
MobileNetV2 (pt)	3.5	1.7	88.70
ResNet-50 (pt)	25.6	4.1	90.32
DenseNet-121 (pt)	8.0	5.7	92.4
ViT (pt)	22.0	8.2	90.38
Swin (scratch)	**1.81**	**1.23**	**94.49**

Furthermore, the lightweight design enables real-time edge deployment, achieving sub-30 ms inference latency on modern smartphones (tested on Snapdragon 778G). This makes the proposed from-scratch Swin Transformer highly suitable for precision agriculture applications where low-power, on-device diagnosis is required.

C. Peak Accuracy Visualization

Fig. 4 compares peak accuracy across models. The Swin Transformer achieves the highest peak accuracy of **94.49%**, outperforming DenseNet (92.40%) and all other baselines.

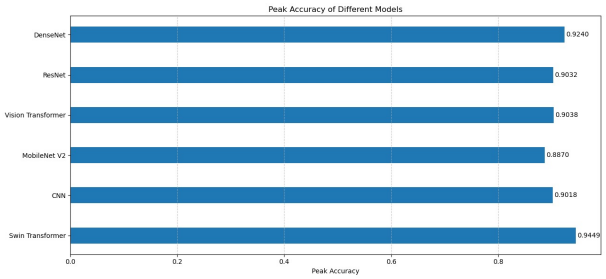


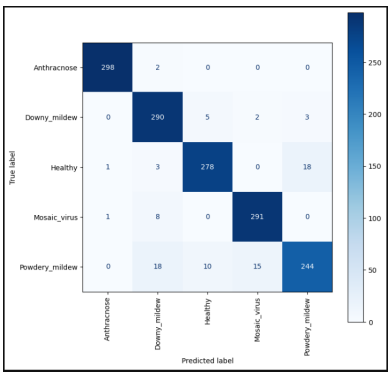
Fig. 4: Peak accuracy comparison among models.

D. Confusion Matrix Analysis

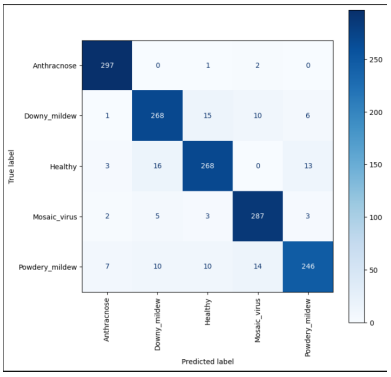
Figure 5 illustrates the confusion matrices for all evaluated models, providing information on their class-wise prediction behavior. Among all architectures, the Swin Transformer exhibits the lowest misclassification rates, particularly in challenging cases involving visually similar classes such as *Powdery mildew* and *Downy mildew*. This indicates the model's superior discriminative capability in fine-grained leaf disease recognition.

E. Model Interpretability

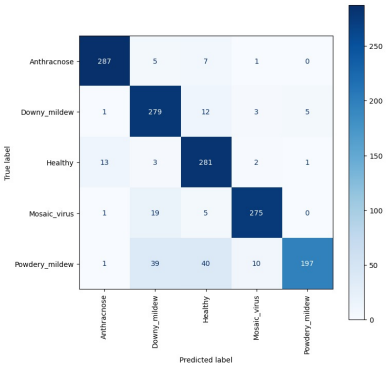
Fig. 6 shows sample detections from the Swin Transformer, demonstrating robustness to lighting, leaf orientation, and background variations. Grad-CAM visualizations in Fig. 7 confirm that the model focuses on disease-affected regions when making predictions.



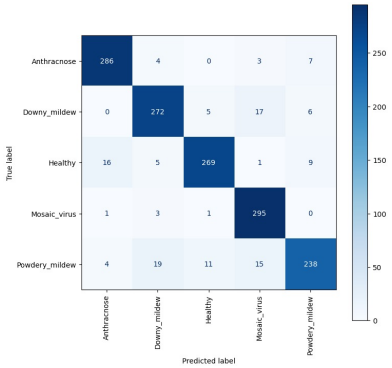
(a) Swin Transformer



(b) Basic CNN

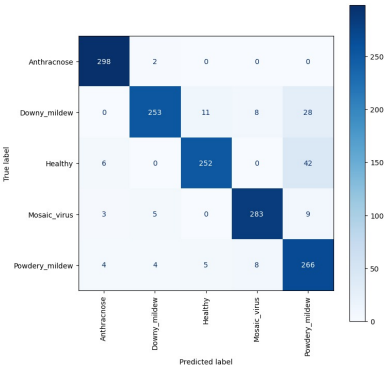


(c) MobileNet V2



(d) Vision Transformer (ViT)

(e) ResNet



(f) DenseNet

Fig. 5: Confusion matrices for all evaluated models, highlighting class-wise prediction performance.

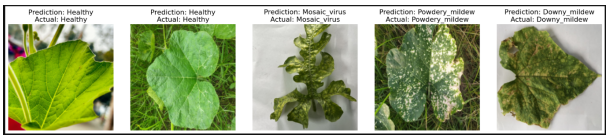


Fig. 6: Sample detections from the Swin Transformer.

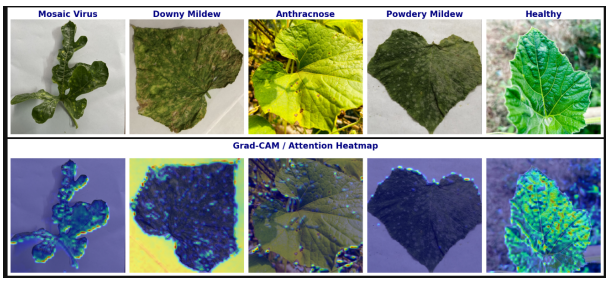


Fig. 7: Grad-CAM visualization for correctly classified samples by the Swin Transformer.

F. Training and Validation Trends

Figure 8 presents the combined loss-over-accuracy and accuracy-over-epochs curves for the Swin Transformer model. The training loss rapidly decreases during the initial epochs, indicating fast convergence. Validation accuracy closely tracks the training accuracy throughout, demonstrating stable learning without significant overfitting. The smooth curves and minimal divergence between training and validation metrics confirm the model's robust generalization capability.

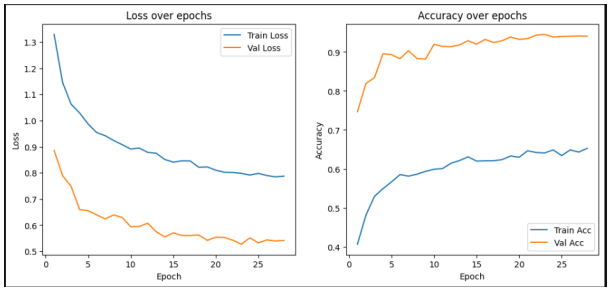


Fig. 8: Training and validation loss and accuracy curves over epochs for the Swin Transformer.

G. Summary of Findings

Overall, the Swin Transformer achieved the highest scores across all evaluation metrics, outperforming all baseline models. It consistently delivered balanced class-wise recognition, minimized errors in visually similar categories, and produced interpretable attention maps that focused precisely on disease-affected regions. These strengths underscore its effectiveness for cucurbit leaf disease detection and its potential for broader agricultural disease recognition tasks.

V. Conclusion

This research conducted a comparative assessment of six deep learning models for detecting diseases in cucurbit leaves, incorporating both convolutional and transformer-based frameworks. The experimental findings revealed that although CNN-based models like ResNet and DenseNet delivered competitive outcomes, the Swin Transformer consistently surpassed all benchmarks, achieving a maximum accuracy of 94.49% and maintaining balanced performance across all categories. Its hierarchical shifted-window approach effectively captured both local and global features, facilitating the reliable identification of visually similar diseases.By utilizing Grad-CAM visualizations, the interpretability was enhanced as it confirmed that the Swin Transformer focused on the disease-specific regions of the leaf. These results suggest that hierarchical vision transformers are promising for developing reliable, efficient, and comprehensible frameworks for disease detection in agriculture.In future research, efforts will be directed towards enlarging the dataset by incorporating additional samples collected from the field, investigating the use of multimodal inputs like weather and soil data, and refining lightweight transformer models for use on mobile and edge devices. By integrating these innovations into precision agriculture systems, farmers could gain access to real-time, user-friendly tools for monitoring crop health and enhancing sustainable yields.

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